Lab 3 – Component Modelling & Architectural Pattern Selection

PS #53 — Freelance Content Creator Escrow Platform

1. Scenario Recap

The platform lets Content Creators submit watermarked draft media deliverables against agreed milestones, and lets Client Sponsors review those drafts and approve them for release. On sponsor sign-off, the system must release the corresponding milestone payment from escrow to the creator. Key constraints from the problem statement: (FR-001) payment must never release without explicit client sign-off, and (NFR-001) uploaded drafts must be automatically watermarked in under 5 seconds, even under peak load, while meeting security requirements.

2. Architectural Style Analysis

Style	Pros (for this scenario)	Cons (for this scenario)
Layered Architecture	Simple to build; clear separation between UI, business logic and data.	Watermarking is CPU-heavy — in a single deployable layer it cannot scale independently, risking the 5s NFR at peak upload volume. A bug in one layer can also affect the whole app.
Microservices Architecture	Each capability (watermarking, escrow, review) can be deployed, secured and scaled independently; a spike in draft uploads only scales the Watermarking Service, not the payment path; failures are isolated.	More operational complexity (service discovery, inter-service calls); needs careful design to keep escrow data consistent across services.
Client-Server Architecture	Centralised control and simple data consistency for a small system.	Single server becomes a bottleneck and single point of failure — risky for a system that must guarantee payment integrity and meet strict latency targets under load.

3. Architecture Selection

“We chose Microservices Architecture for the Freelance Content Creator Escrow Platform.”

Reason 1 — Independent scaling for a compute-heavy NFR:

Automated watermarking (NFR-001) is CPU/GPU-intensive and must complete in under 5 seconds even during upload bursts. As an isolated Watermarking Service, it can be horizontally auto-scaled on its own infrastructure without over-provisioning the rest of the system, and without the Escrow/Payment path competing for the same compute resources.

Reason 2 — Isolation of trust boundaries:

Milestone payments (FR-001) require funds to move only after an explicit sponsor sign-off. Keeping Escrow & Payment as its own service means the money-movement logic is decoupled from higher-churn, higher-exposure services such as media upload and watermarking — each service can evolve and be redeployed independently without risking the integrity of the payment logic.

Security Advantage:

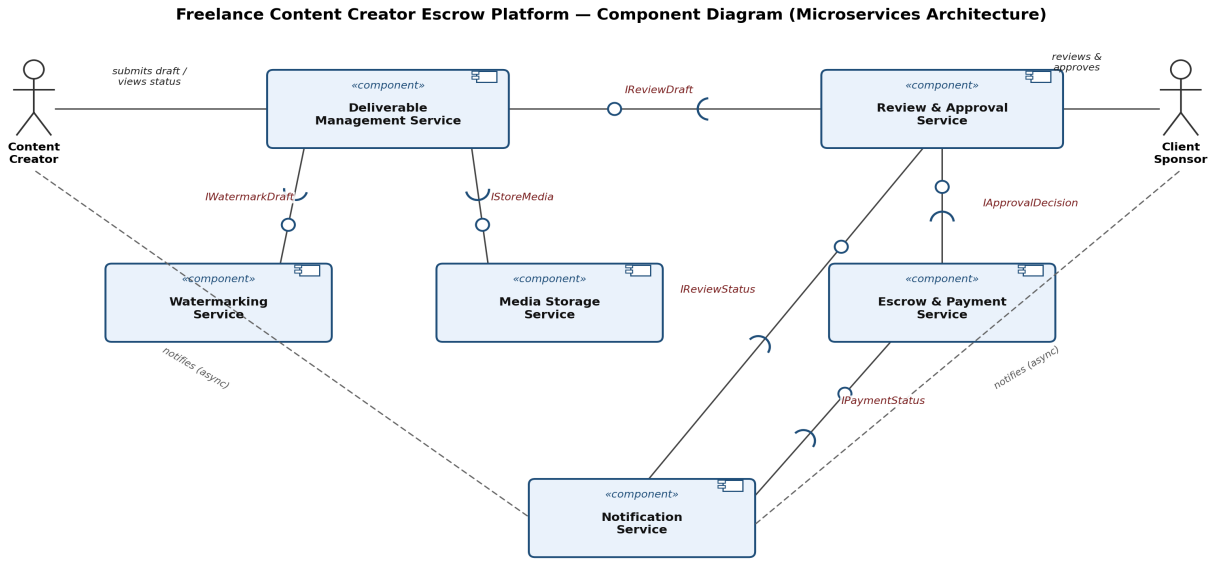
Because each capability is its own service, the Escrow & Payment Service can sit behind stricter network segmentation and authentication (e.g. only accessible through an internal API gateway with least-privilege, service-to-service credentials) than the public-facing upload/watermarking path. A compromise of the media or watermarking service therefore does not directly expose the payment/escrow data — the blast radius of any single breach is contained to that one service.

Performance Benefit:

The Watermarking Service can be scaled out independently (multiple instances behind a load balancer) purely in response to draft-upload traffic, which is the only realistic way to consistently hit the sub-5-second watermarking target under peak load without wastefully scaling the entire application.

4. UML Component Diagram

Six components, six labelled interfaces (provided = ball / required = socket notation), and the two primary actors (Content Creator, Client Sponsor) are shown below.



Component	Responsibility
Deliverable Management Service	Accepts draft submissions/milestones from the Creator; orchestrates watermarking and storage; exposes review data.
Watermarking Service	Applies automated watermark to draft media within the 5s NFR; independently scalable.
Media Storage Service	Persists draft and final media assets.
Review & Approval Service	Presents drafts to the Client Sponsor and captures the approval sign-off decision.
Escrow & Payment Service	Holds milestone funds in escrow and releases payment only on a confirmed approval decision.
Notification Service	Pushes async status updates (review status, payment status) to both actors.